Physics 9th – Chapter-2-Kinematics (Short Response Questions)

Q1. In a park, children are enjoying a ride on Ferris wheel. What kind of motion the big wheel has and what kind of motion the riders have?

Ans:

- The big wheel rotates about a fixed axis passing through its centre, so it has rotatory motion.
- The riders move along a circular path around the centre while their seats stay upright (no change in orientation), so they have circular motion, a type of translatory motion.
- Concept: in rotatory motion the object turns about its own axis; in circular motion the object moves along a circular path.

Q2. A boy moves for some time, give two situations in which his displacement is zero but covered distance is not zero?

Ans:

- He completes one full round of a 400 m circular track: distance = 400 m, displacement = 0.
- He walks from home to school and comes back home: distance = twice the path length, displacement = 0.
- Concept: displacement depends only on initial and final positions; distance is the length of the path travelled.

Q3. A stone tied to string is whirling in circle, what is direction of its velocity at any instant?

Ans:

- The velocity at any instant is along the tangent to the circle at that point, perpendicular to the string (radius).
- If the string breaks, the stone flies off along this tangent.
- Concept: its speed may be constant but its direction changes continuously, so its velocity is variable.

Q4. Is it possible to accelerate an object without speeding it up or slowing it down?

Ans:

- Yes. Acceleration is the rate of change of velocity, and velocity can change by changing direction only.
- Example: a stone whirled in a circle or a satellite moving with constant speed in a circular orbit around the Earth.
- Concept: velocity is a vector, so a change in direction alone produces acceleration.

Q5. Can a car moving towards right have direction of acceleration towards left?

Ans:

- Yes, when the car moving towards right is slowing down (brakes are applied).
- Its velocity is towards right but its acceleration is towards left, opposite to the motion.
- Concept: acceleration opposite to velocity is called deceleration (negative acceleration).

Q6. With the help of daily life examples, describe the situations in which: (a) acceleration is in the direction of motion (b) acceleration is against the direction of motion (c) acceleration is zero and body is in motion.

Ans:

- (a) A car speeding up when the signal turns green; a ball dropped from a roof (velocity increases downward).
- (b) A car or bicycle slowing down when brakes are applied; a ball thrown vertically upward.
- (c) A car moving at a constant 100 km/h on a straight motorway; a train moving with uniform velocity on a straight track.
- Concept: acceleration along motion increases speed, against motion decreases speed, zero acceleration means uniform velocity.

Q7. Examine distance-time graph of a motorcyclist (as shown), what does this graph tell us about the speed of motorcyclist? Also plot its velocity-time graph.

Ans:

- The graph is a straight line sloping upward, so the motorcyclist covers equal distances in equal intervals of time.
- Its slope ($\Delta s/\Delta t$) is constant, so the motorcyclist moves with uniform speed (about 8 m/s from the graph) and zero acceleration.
- Its velocity-time graph is a horizontal straight line parallel to the time axis at this constant speed.
- Concept: gradient of distance-time graph gives speed.

Q8. Which controls in the car can produce acceleration or deceleration in it?

Ans:

- Accelerator (gas pedal) increases the speed, producing acceleration.
- Brake pedal decreases the speed, producing deceleration.
- Steering wheel changes the direction of motion, which also produces acceleration (change in velocity).

Q9. If two stones of 10 kg and 1 kg are dropped from a 1 km high tower. Which will hit the ground with greater velocity? Which will hit the ground first? (Neglect the air resistance)

Ans:

- Both stones fall with the same acceleration $g = 9.8\ m/s^2$, which does not depend on mass.
- So both hit the ground at the same time (about 14.3 s) and with the same velocity (about 140 m/s).
- Concept: all freely falling bodies have the same acceleration due to gravity (Galileo).

Q10. A 100 g ball is just released (from rest) and another is thrown downward with velocity of 10 m/s, which will have greater acceleration? (Neglect the air resistance)

Ans:

- Both balls have the same acceleration $g = 9.8\ m/s^2$ directed downward.
- The thrown ball has a greater velocity at every instant, but its rate of change of velocity is the same.

- Concept: acceleration due to gravity is independent of initial velocity and mass.